

Eye Kalman Project

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Overview

1. Context
2. Objectives
3. Laser 101
4. Mathematical Model
5. Discretization
6. Simulations



Context

The Human Eye

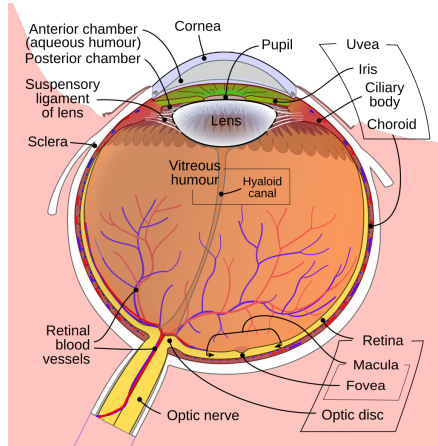


Figure 1: *[Schematic diagram of the human eye. Wikipedia]*

Light-Tissue Interactions



Coefficients of interest: **absorption** $\mu_{a,i}(\lambda)$, **scattering** $\mu_{s,i}(\lambda)$.

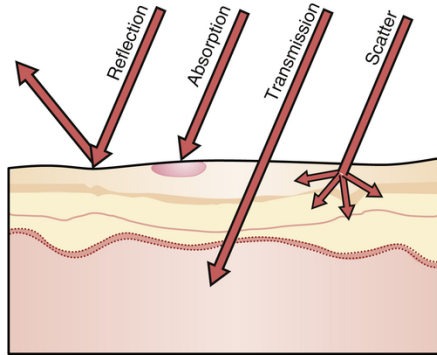


Figure 2: Laser tissue interactions. [*"Laser Tissue Interactions: Biological Factors to Consider for Dermatology [Blo19]."*]

The background of the slide is composed of two large, overlapping geometric shapes. A teal-colored shape occupies the top-left corner, while a light beige shape occupies the bottom-left corner. The rest of the slide is white. The word "Objectives" is centered in the white area.

Objectives



Project Objectives

- ▶ **Develop a mathematical model** coupling:
 - ▶ **Fluence rate** Φ [W/m^2]: total radiant power per unit area from all directions.
 - ▶ **Temperature** T [K or $^{\circ}\text{C}$]: thermal response in tissue.
- ▶ Build a realistic **3D geometric model** of the eye.
- ▶ Solve the **direct problem**:
Predict **temperature distribution** from known optical properties.
- ▶ Tackle the **inverse problem** using the **Ensemble Kalman Filter (EnKF)**:
Estimate tissue optical properties:
 - ▶ $\mu_a(\lambda)$: absorption coefficient
 - ▶ $\mu_s(\lambda)$: scattering coefficient
- ▶ **Visualize** and **analyze** the spatiotemporal results.
- ▶ Conduct a **sensitivity analysis** to identify dominant parameters.

The background of the slide is composed of three geometric sections. A teal-colored triangle is in the top-left corner. A light gray triangle is in the bottom-left corner. The remaining area is a white trapezoid. The text 'Laser 101' is centered within the white section.

Laser 101

Laser Types

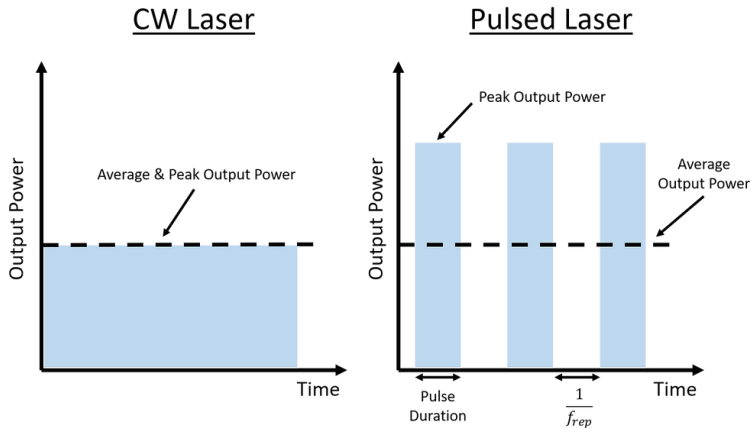


Figure 3: Continuous Wave (CW) vs Pulsed lasers.

[<https://www.laser-protect.fr/en/post/laser-operating-modes-continuous-and-pulsed>]



Key Parameters

- ▶ **Pulse Duration** τ : Ranges from fs (ultrashort) to ms (long-pulse).
- ▶ **Wavelength** λ : Distance between wave peaks.
Optical properties depend on λ : $\mu_a(\lambda)$ = absorption, $\mu_s(\lambda)$ = scattering.
- ▶ **Laser Power** P : Typically 0.1 W–2 W.
- ▶ **Beam Waist** w_0 : The tightest focus of the beam — where intensity peaks.
- ▶ **Beam Divergence** θ : Angular spread of the beam, in rad or mrad. For Gaussian beams:

$$\theta = \frac{\lambda}{\pi w_0}$$

- ▶ **Rayleigh Range** z_R : Distance from the waist to where beam cross-section doubles:

$$z_R = \frac{\pi w_0^2}{\lambda}$$

Beam Profiles

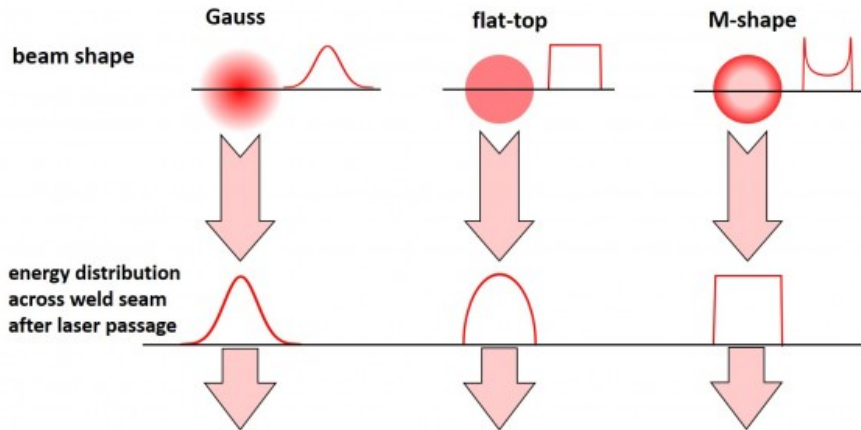
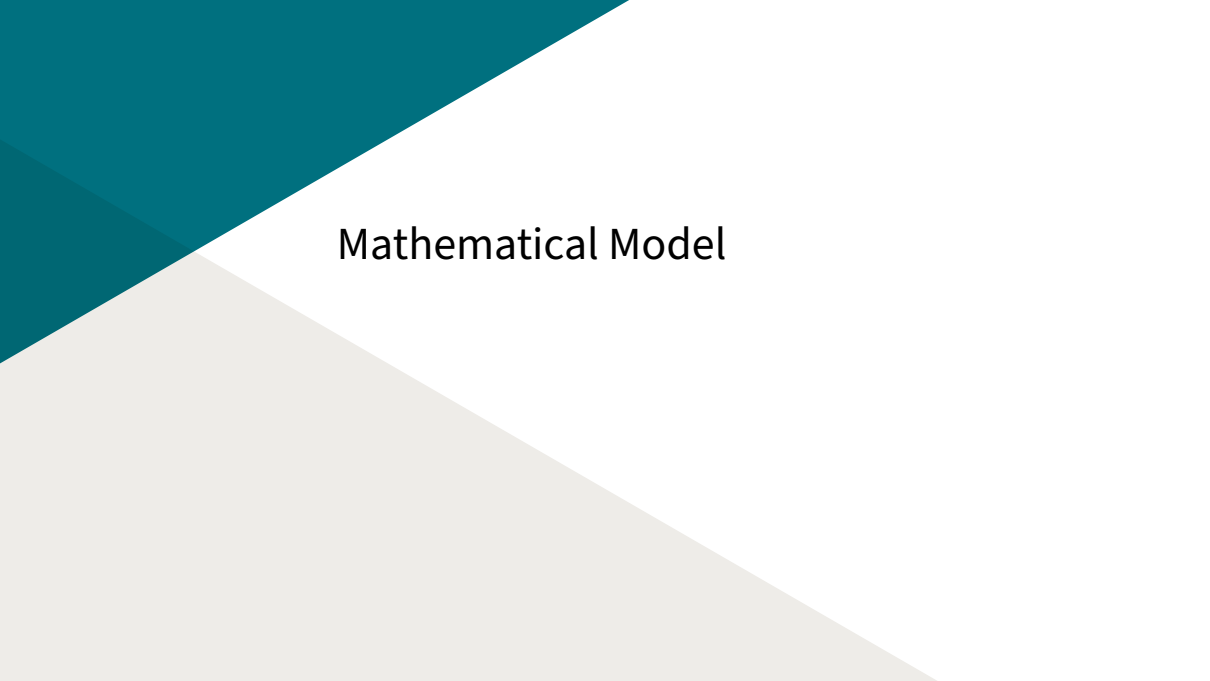


Figure 4: Different beam profiles. ["Beam Profiles" by Probylas]

The background of the slide is composed of two large, overlapping geometric shapes. A teal-colored shape occupies the top-left corner, while a light gray shape occupies the bottom-left corner. The rest of the slide is white. The text 'Mathematical Model' is centered in the white area.

Mathematical Model

Eye Geometry $\Omega = \bigcup_{i=1}^{10} \Omega_i$

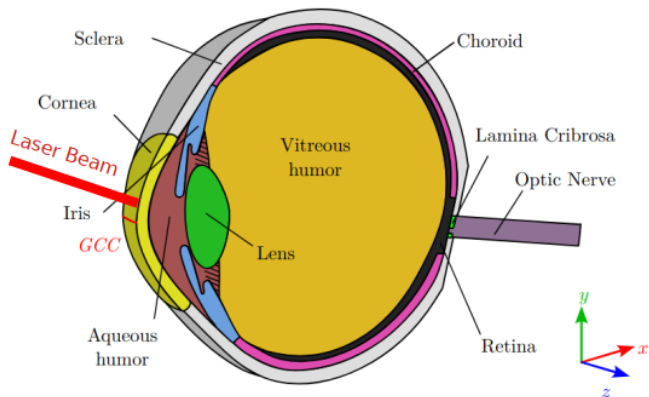


Figure 5: Vertical cut of the geometrical model of the human eye. [*"Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball". Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].*]

Eye Geometry: Boundary Conditions

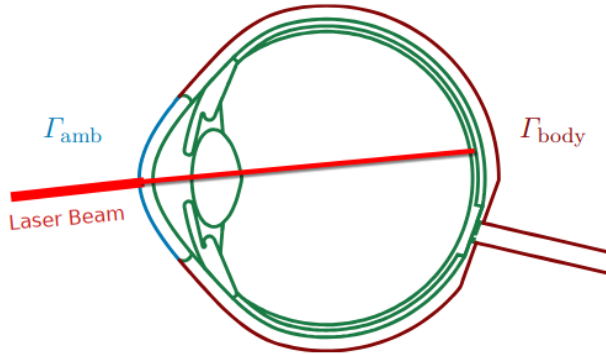


Figure 6: Vertical cut of the geometrical model of the human eye. [*"Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball". Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].*]

The Pennes Bioheat Equation



$$\left\{ \begin{array}{ll} \rho_i c_{p_i} \frac{\partial T_i}{\partial t} - \nabla \cdot (k_i \nabla T_i) = Q_{l,i} + Q_{m,i} + Q_{p,i} & \text{on } \Omega = \bigcup_{i=1}^{10} \Omega_i, \\ -k_i \nabla T_i \cdot n_i = \underbrace{h_{\text{amb}}(T_i - T_{\text{amb}})}_{(i)} + \underbrace{\sigma \varepsilon (T_i^4 - T_{\text{amb}}^4)}_{(ii)} + \underbrace{E}_{(iii)} & \text{on } \Gamma_{\text{amb}}, \\ -k_i \nabla T_i \cdot n_i = h_{\text{bl}}(T_i - T_{\text{bl}}) & \text{on } \Gamma_{\text{body}}, \\ T_i = T_j & \text{on } \Omega_i \cap \Omega_j, \\ k_i \nabla T_i \cdot n_i = -k_j \nabla T_j \cdot n_j & \text{on } \Omega_i \cap \Omega_j. \end{array} \right.$$

$Q_{l,i} = \mu_{a,i}(\lambda) \times \Phi_{\lambda}(x, y, z)$: **laser** heat generation,

$Q_{m,i}$: **metabolic** heat generation $\rightarrow$ ignored since negligible compared to the others,

$Q_{p,i} = \rho_{\text{bl}} \omega_{\text{bl}} c_{p_{\text{bl}}} (T_{\text{bl}} - T_i)$: blood **perfusion** heat generation.

The Radiative Transfer Equation (RTE)



Describes the propagation of light in a medium.

$$\left(\frac{1}{c} \frac{\partial}{\partial t} + \boxed{s \cdot \nabla} + \boxed{\mu_a(r) + \mu_s(r)} \right) L(\mathbf{r}, \mathbf{s}, t) = \boxed{\mu_s(r) \int_{4\pi} L(r, \mathbf{s}', t) \beta(\mathbf{s}, \mathbf{s}') d\mathbf{s}'} + \boxed{Q(r, \mathbf{s}, t)},$$

$L(r, \mathbf{s}, t)$: **radiance** at position $\mathbf{r}$, in direction $\mathbf{s}$, $(r, \mathbf{s}) \in \mathbb{R}^3 \times \mathbb{S}^2$

c : speed of light [ms^{-1}],

$\beta(\mathbf{s}, \mathbf{s}')$: phase function [dimensionless].

$\boxed{s \cdot \nabla}$: **advection** — directional transport of light,

$\boxed{\mu_a(r) + \mu_s(r)}$: **attenuation** due to absorption and scattering,

$\boxed{\mu_s(r) \int \dots}$: **in-scattering** contribution,

$\boxed{Q(r, \mathbf{s}, t)}$: **source** term.



The Diffusion Approximation

Valid when $\mu_a \ll \mu_s$.

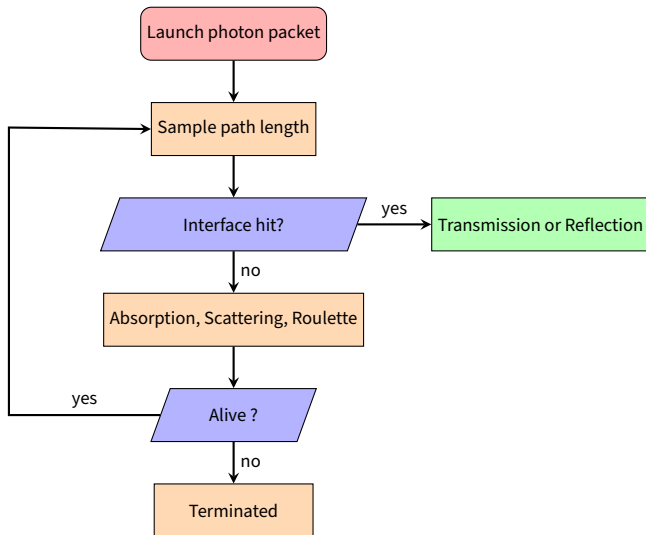
$$\left\{ \begin{array}{ll} \frac{1}{c} \frac{\partial \Phi_i}{\partial t} - \nabla \cdot (D_i \nabla \Phi_i) + \mu_{a,i}(\lambda) \Phi_i = S & \text{on } \Omega = \bigcup_{i=1}^{10} \Omega_i, \\ \Phi_i + 2 \alpha_i D_i \nabla \Phi_i \cdot n_i = \Phi_{\text{amb}} & \text{on } \Gamma_{\text{amb}}, \\ \Phi_i = \Phi_j & \text{on } \Omega_i \cap \Omega_j. \end{array} \right.$$

$D_i = \frac{1}{3(\mu_{a,i}(\lambda) + \mu'_{s,i}(\lambda))}$: diffusion coefficient,

$\alpha_i = \frac{1 + R_i}{1 - R_i} \in [0, 1]$: albedo (*i.e.* fraction of light scattered back),

R_i : reflection coefficient.

The MC Approach (RTE-MC)





RTE-MC: Beer-Lambert Law

Using the **Beer-Lambert law**, for a **homogeneous medium** of length x we have:

$$N_x = N_0 e^{-\mu_t x}$$

Where:

- ▶ **N_0** : initial number of photons
- ▶ **N_x** : number of photons after traveling distance x
- ▶ **$\mu_t = \mu_a + \mu_s$** : **attenuation coefficient**, sum of absorption μ_a and scattering μ_s



RTE-MC: Path Length Distribution

The probability density function (PDF) of the photon **path length** x in a medium is:

$$f_x(x) = \underbrace{\mu_t}_{\text{attenuation coeff.}} e^{-\mu_t x} \mathbb{1}_{\{x \geq 0\}}$$

The cumulative distribution function (CDF) is given by:

$$F_x(x) = \int_{-\infty}^x f_x(t) dt = \int_0^x \mu_t e^{-\mu_t t} dt$$

Evaluating the integral:

$$F_x(x) = \mu_t \left[-\frac{1}{\mu_t} e^{-\mu_t t} \right]_0^x = 1 - e^{-\mu_t x} \mathbb{1}_{\{x \geq 0\}}$$

Note: $F_x(x)$ is strictly increasing on $\mathbb{R}^+$ and hence **invertible**, allowing us to sample path lengths from uniform random variables.



RTE-MC: Inverse Transform Sampling

We want to find the **inverse** of the CDF to sample **path length x** from a **uniform random variable $\xi \sim U(0, 1)$** .

$$\begin{aligned}y &= 1 - e^{-\mu_t x}, \quad y \in (0, 1) \\e^{-\mu_t x} &= 1 - y \\-\mu_t x &= \ln(1 - y)\end{aligned}$$

Since $y \in (0, 1)$, define $\xi := 1 - y \in (0, 1)$. Then

$$s := x = -\frac{\ln(\xi)}{\mu_t}$$



RTE-MC: Anisotropic Scattering

When the medium is **anisotropic**, the photon travels a distance s in a random direction θ and then scatters.

To sample the **scattering angle** θ , we use the **Henye-Greenstein** phase function:

$$\cos(\theta) = \begin{cases} 2\xi - 1 & \text{if } g \approx 0 \\ \frac{1 + g^2 - \left(\frac{1-g^2}{1-g+2g\xi}\right)^2}{2g} & \text{otherwise} \end{cases}$$

Here, $g \in (-1, 1)$ is the **anisotropy factor**, representing the degree of **forward scattering**.



RTE-MC: Energy Deposition

At each space step s , the photon (or photon packet) deposits part of its **energy w** (initialized at 1) into the element it lands in.

$$w \leftarrow w \cdot \frac{\mu_a}{\mu_t}$$

Where w is the **updated energy** of the photon packet after absorption in the current element.

Note: Intermediate elements traversed between steps may not receive any energy deposition unless explicitly handled.

RTE-MC: Absorption Map

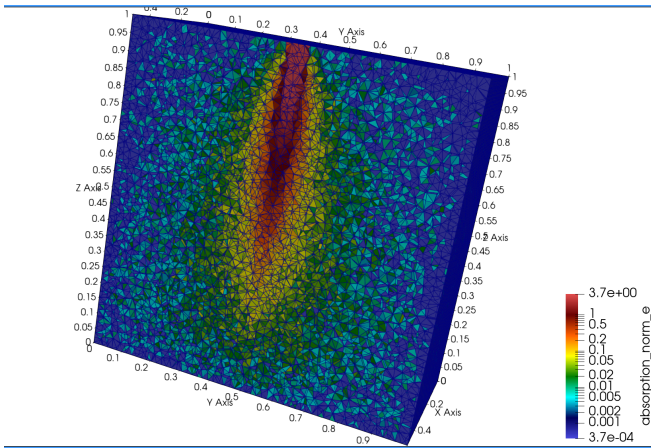


Figure 7: Early results of the map of absorption on a cube.

Bounding Volume Hierarchy (BVH)

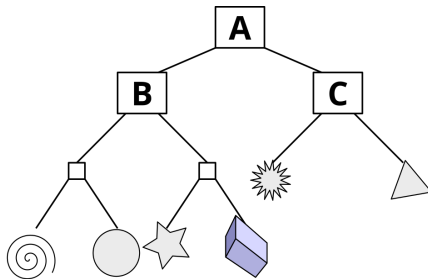
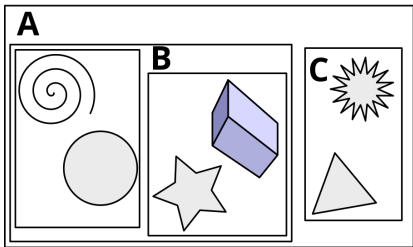


Figure 8: Example of bounding volume hierarchy (BVH) in two dimensions. *[Wikipedia]*

RTE-MC: Fresnel Illustration

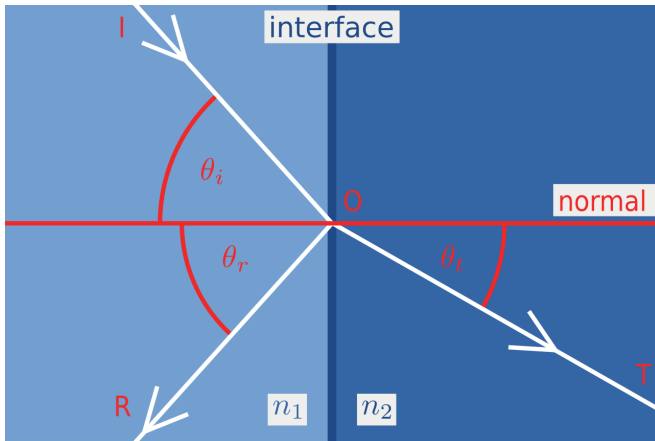


Figure 9: Fresnel equations illustrated ($\theta_i = \theta_r$). ["Fresnel equations" Wikipedia]



RTE-MC: Interface Interactions

When a photon hits an interface Ω_{ij} :

- ▶ **Snell's law** gives the **refracted direction**:

$$n_1 \sin(\theta_i) = n_2 \sin(\theta_t)$$

- ▶ **Fresnel equations** (or Schlick's approximation) give the **reflection probability**:

$$R(\theta_i) \approx R_0 + (1 - R_0)(1 - \cos \theta_i)^5$$

$$\text{with } R_0 = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2.$$

Algorithm:

- ▶ Draw $u \sim \mathcal{U}(0, 1)$.
- ▶ If $u < R(\theta_i)$: **reflect**, else **refract**.



RTE-MC: Russian Roulette

We expect the photon's energy to **decrease rapidly**. To avoid tracking all **low-energy photons** while maintaining physical accuracy, we use the **Russian Roulette** technique. Let $p \in (0, 1)$ be the survival probability. If $w < w_{\min}$, then:

Russian Roulette:

Sample a realisation $\xi \sim \mathcal{U}(0, 1)$

if $\xi < p$ **then**

Photon survives: $w \leftarrow \frac{w}{p}$

else

Photon is terminated

end if



Pennes Bioheat Equation & RTE-MC Coupling

- ▶ The **RTE-MC** provides the **normalized absorption map A**.
- ▶ The **source term Q_l** in the **Pennes equation** is given by:

$$Q_l(x, y, z) = P \times A$$

where **P** is the **laser power** and **A** is the **absorption map** from RTE-MC.

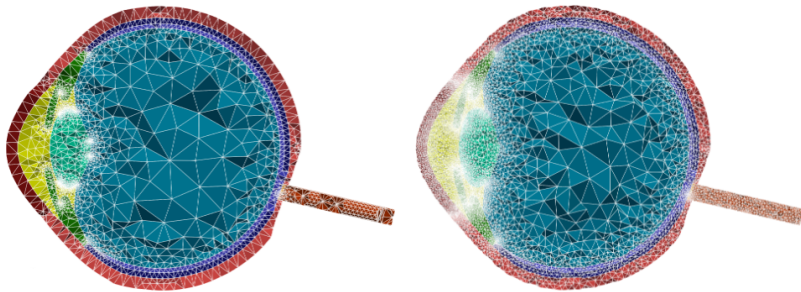
- ▶ Solving the **Pennes equation** yields the **temperature distribution $T(x, y, z)$** within the eye.

This coupling allows simulation of light-induced heating effects with spatial precision.

The background of the slide is composed of two large, overlapping geometric shapes. A teal-colored shape occupies the top-left corner, while a light gray shape occupies the bottom-left corner. The rest of the slide is white. The word "Discretization" is centered in the white area.

Discretization

Space Discretization



(a) Original mesh generated by Salome, with $4.64 \cdot 10^5$ tetrahedrons: $h_{\min} = 5.77 \cdot 10^{-6}$, $h_{\max} = 5.76 \cdot 10^{-3}$.
(b) Mesh refined around the anterior and posterior chambers, with $9.4 \cdot 10^5$ elements: $h_{\min} = 5.09 \cdot 10^{-5}$, $h_{\max} = 3.12 \cdot 10^{-3}$.

Figure 10: Meshed geometry of the eye, over a vertical plane. Characteristic sizes are given in meters. [*"Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball"*. Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].]

Space Discretization: Test Case

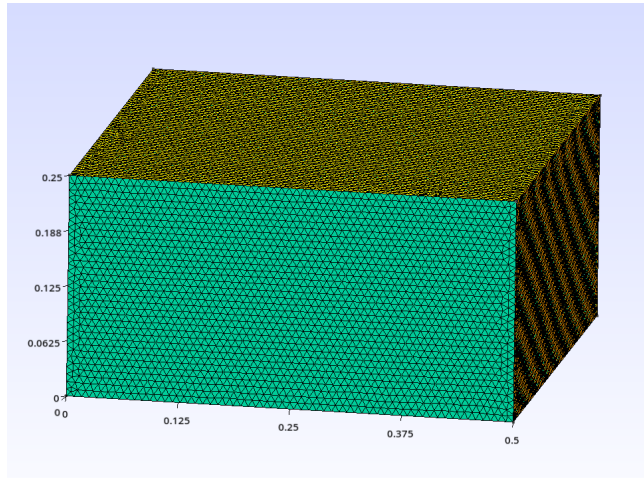


Figure 11: 0.5 x 0.5 x 0.25 cm single layer cube with 81907 nodes and 495013 elements.

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Simulations



RTE-MC Simulation Parameters

- ▶ Beam Diameter: 0.2 cm
- ▶ Absorption Coefficient μ_a : 5 cm⁻¹
- ▶ Scattering Coefficient μ_s : 100 cm⁻¹
- ▶ Anisotropy Factor g : 0.9
- ▶ Number of Photons: 2,000,000

RTE-MC: Absorption Map

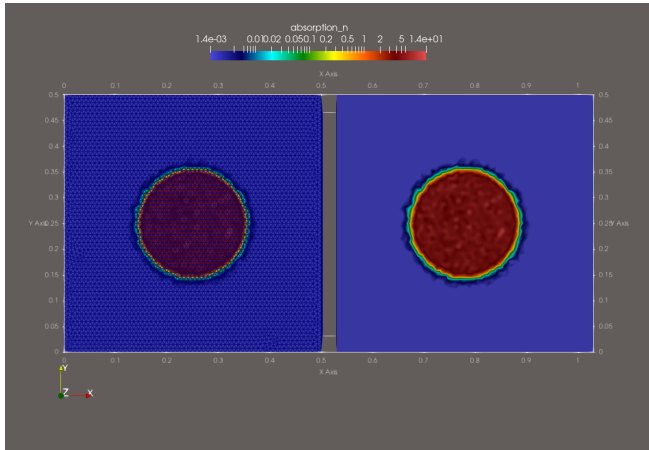


Figure 12: Log scale top view of the absorption map.

RTE-MC: Absorption Map

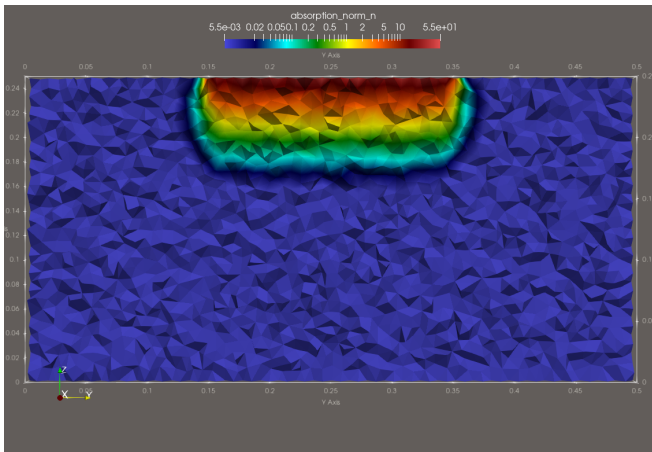


Figure 13: Log scale clipped view of the absorption map for the cube geometry.



Heat Simulation Parameters

- ▶ Laser Power P : 0.5 W
- ▶ Simulation Time Step Δt : 0.01 s
- ▶ Total Simulation Time T : 15 s
- ▶ Time Scheme: Crank-Nicolson
- ▶ Boundary Conditions: Homogeneous, Neumann on top, Dirichlet on others
- ▶ Initial Temperature T_0 : 0 °C

Heat Simulation Results

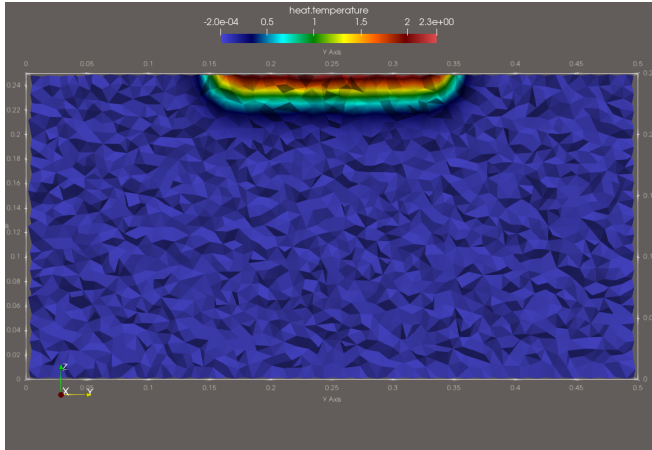


Figure 14: $t=0.2$ s

Heat Simulation Results

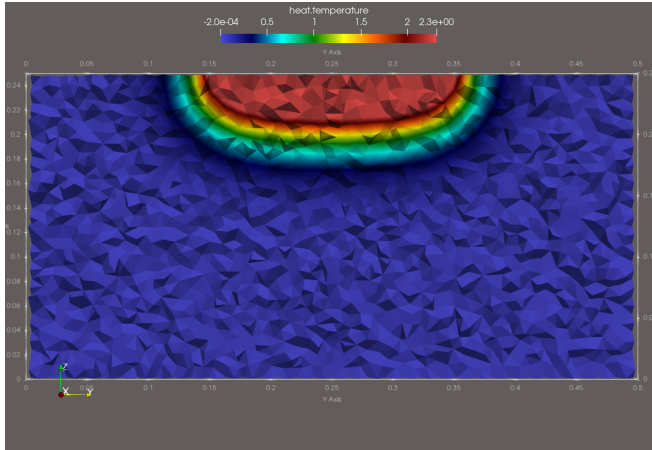


Figure 15: $t=1$ s

Heat Simulation Results

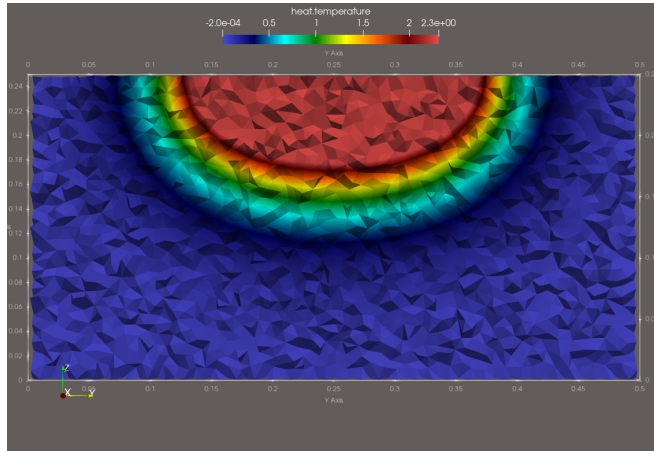


Figure 16: $t=3$ s

Heat Simulation Results

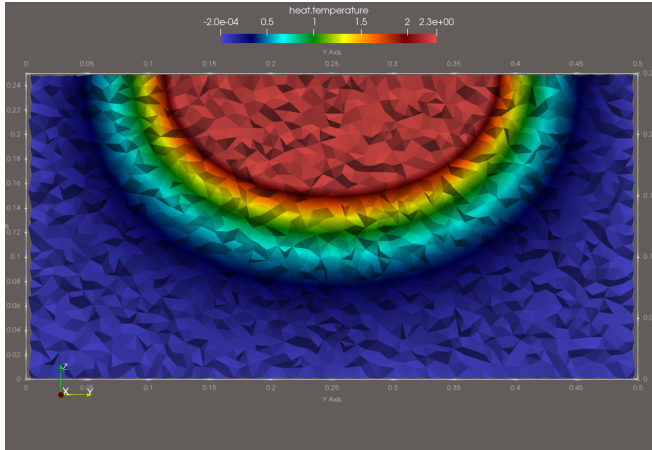


Figure 17: $t=5$ s

Heat Simulation Results

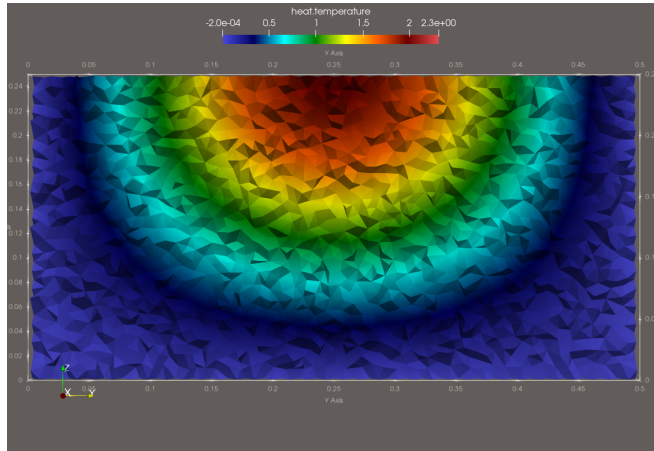


Figure 18: $t=10$ s

Heat Simulation Results

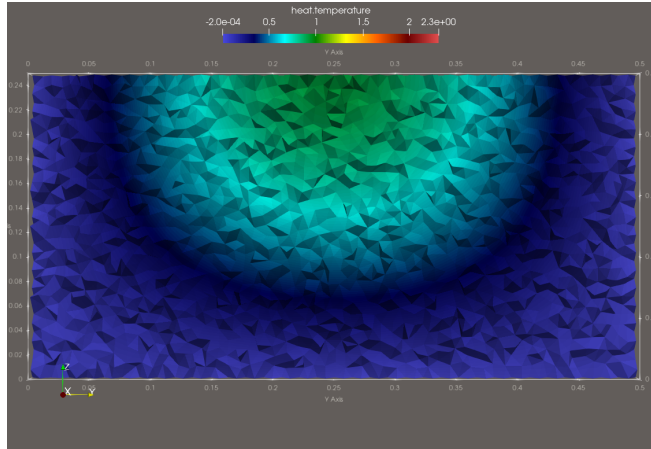


Figure 19: $t=15$ s

The end



Thank you for your attention!



Any questions?

Contact Information:

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References

[Blo19] FindLight Blog. *Laser Tissue Interactions: Biological Factors to Consider for Dermatology*. 2019. URL:
<https://www.findlight.net/blog/laser-tissue-interactions/>.



[SPS24] Thomas Saigre, Christophe Prud'homme, and Marcela Szopos. "Model Order Reduction and Sensitivity Analysis for Complex Heat Transfer Simulations inside the Human Eyeball". In: *International Journal for Numerical Methods in Biomedical Engineering* 40.11 (Sept. 9, 2024), e3864. ISSN: 2040-7939, 2040-7947. DOI: 10.1002/cnm.3864. URL:
<https://onlinelibrary.wiley.com/doi/10.1002/cnm.3864>.

Heat Simulation Validation

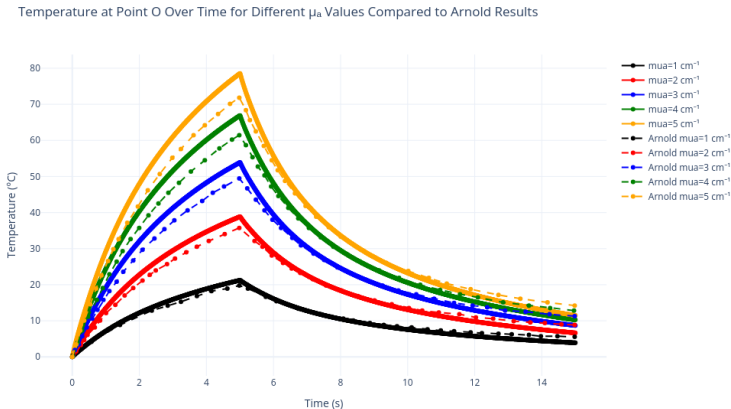


Figure 20: Validation of heat simulation results.

Heat Simulation Validation



Temperature at Point O Over Time for Different μ_a Values Compared to Arnold Results

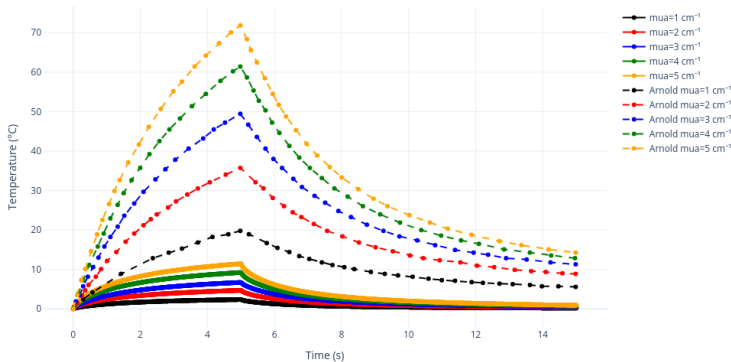


Figure 21: New algo results vs Arnold.